Ho Chi Minh City University of Technology (HCMUT), VNU-HCM

Faculty of Computer Science and Engineering

Course Project Handbook

Assignments 1–3

Deep Learning and Its Applications

Course code: CO3133 · Semester-261

Instructor: Lê Thành Sách

Document type: Unified assignment handbook for students

Part I: course-wide rules · Parts II–IV: assignment specifications

Contents

Part I

Course-Wide Requirements

1 Course project overview

This handbook defines the three course projects for Deep Learning and Its Applications (CO3133), Semester-261:

1. **Assignment 1 — Foundations of Deep Learning Pipelines and Architectures**
   Topic: From Linear Models to Modern Sequence Models: A Comparative Study for Image Classification.
2. **Assignment 2 — Deep Learning on Large-Scale Data and Specialized Tasks**
   Topic: Deep Learning on Large-Scale Data for Specialized Computer Vision or Language Tasks.
3. **Assignment 3 — Multimodal Deep Learning**
   Topic: Multimodal Deep Learning: Representation, Fusion, and Evaluation.

Part I states rules that apply to all three assignments. Parts II–IV state assignment-specific requirements. When a Part II–IV rule conflicts with a general statement, the assignment-specific rule takes precedence for that assignment only.

1.1 Group format

- Work in **groups of 3–4 students**.
- Register the group on the course LMS page: `https://lms.hcmut.edu.vn/course/view.php?id=142848` (follow the group-registration activity posted there).
- The same registered group completes **all three** assignments.

1.2 Course project grade weights

The three assignments together constitute the **course project** component. Their relative weights are:

Assignment	Focus	Weight
Assignment 1	Foundations / pipelines & architectures	40%
Assignment 2	Large-scale data & specialized tasks	30%
Assignment 3	Multimodal deep learning	30%
	Total (course project)	**100%**

- These weights apply to the **course project / BTL** component. They do not redefine other course components (quizzes, midterm, final exam, etc.), if any, which are published separately on the LMS.
- Each assignment is first graded on a 100-point scale using its *within-assignment* rubric (Parts II–IV). That score is then multiplied by the weight above.
- Example: an Assignment 1 score of 85 contributes $0.40 \times 85 = 34$ points toward the course-project total (out of 100).
- Core mandatory work dominates each assignment grade. Extensions cannot replace missing core requirements.

1.3 Submission milestones

All assignments follow milestone deadlines (Draft/Proposal/Final) due at **23:59 Asia/Ho Chi Minh (GMT+7)** on the dates in Section 7. Groups must register and publish a GitHub Pages skeleton by **09 Sep 2026, 23:59 GMT+7**. Late submissions follow Section 7.2.

1.4 Shared principles

- Required and optional work must be clearly distinguished.
- Vague claims such as "large enough dataset", "good results", or "sufficient analysis" without quantitative criteria are not acceptable.
- Accuracy alone is not a sufficient basis for declaring one method better than another.
- High numbers without explanation or reproducibility do not receive full credit.

2 Course website and presentation requirements

Every group must maintain a public website on **GitHub Pages** covering all course assignments.

2.1 Shared landing page

The landing page must include:

Institutional information.

- Ho Chi Minh City University of Technology – VNU-HCM.
- Faculty of Computer Science and Engineering.

Course information.

- Course name: Deep Learning and Its Applications.
- Course code: CO3133.
- Semester-261.
- Instructor: Lê Thành Sách.

Group information.

- Group name or group ID.
- Full name of each member.
- Student ID of each member.
- Main role or contribution of each member.
- Link to each member's GitHub profile, *if available* (do not invent fake links).
- Link to the group's code repository.

Assignment index. Clear links to Assignment 1, Assignment 2, and Assignment 3 pages.

AI Usage Disclosure (landing). A shared disclosure section summarizing AI use across the course projects (details in Section 5).

2.2 Assignment page (minimum content)

Each assignment page must include:

- Assignment title.
- Group members.
- Instructor.
- Problem statement.
- Dataset description and EDA.
- Methodology.
- Experimental setup.

- Results.
- Comparison and discussion.
- Error analysis.
- Limitations and conclusion.
- Link to source code.
- Link to checkpoint(s) or instructions to reproduce checkpoints.
- Link to report/slides.
- Link to the YouTube presentation video.
- Assignment-specific AI Usage Disclosure.

2.3 YouTube presentation video (mandatory)

- Every assignment requires a YouTube video.
- Unified title format: `CO3133-Semester-261 - Group [ID] - Assignment [1/2/3]`.
- All members must present or defend part of the content.
- Required coverage: problem and data; method; experimental setup; main results; error analysis; conclusion; short demo if appropriate.
- Assignment 3 should also discuss modality interaction (support vs. conflict cases).
- Do not merely read slides aloud.
- Visibility must be **Public** or **Unlisted**.
- The group is responsible for verifying the link before submission.

3 Report requirements

Each assignment report must contain three main parts. Vague claims without evidence are not acceptable. Assignment-specific EDA and method expectations appear in Parts II–IV.

3.1 Part 1 — Problem and Data Description

Problem formulation. Real-world motivation; input; output; prediction unit; mathematical formulation when appropriate; main difficulties.

Dataset description. Source, version, license; collection/labeling process if known; sample counts and annotations; class or target distribution; storage size and formats; bias and limitations.

EDA. Task-appropriate exploratory analysis (see the assignment chapter for required EDA items).

Data splitting. Train/validation/test; official or custom split; seed; stratification; split unit; leakage prevention; subset selection rules (if any).

Preprocessing. Cleaning; resize/normalization/tokenization/padding; augmentation; invalid annotation handling; input tensor shape(s); one example batch after preprocessing.

3.2 Part 2 — Methodology

Pipeline overview. Include a diagram of the form:

```
Raw data → preprocessing → data loader → model → loss
→ optimization → prediction → post-processing → evaluation
```

Multimodal projects may replace the middle stages with encoders and fusion as appropriate.

Baseline and main methods. Describe architecture, input representation, loss, metrics, strengths and limits, and parameter count. State which parts were self-implemented and which came from libraries (name the libraries).

Training setup. Optimizer; learning rate; batch size; epochs; scheduler; regularization; early stopping; checkpoint criterion; seed; hardware; mixed precision (if used); training time; hyperparameter selection method.

Experimental design. For each experiment: hypothesis; changed factor; fixed factors; metrics; conclusion criteria.

3.3 Part 3 — Implementation Results

Training behavior. Loss/metric curves; selected checkpoint; overfitting/underfitting; training time; implementation issues and fixes.

Quantitative results. Primary and secondary metrics; parameter counts; training/inference time; configurations; mean ± standard deviation over multiple runs when feasible.

Qualitative results. Correct predictions; difficult but correct cases; failures; predictions vs. ground truth; task-appropriate visualizations.

Comparison and discussion. Which method is better and by how much; where differences appear; causes related to data/architecture/loss/training; accuracy–speed–parameter trade-offs; whether hypotheses are supported; conclusions that cannot yet be drawn. Listing metrics and declaring the highest score as best is not sufficient.

Error analysis. For each error group: description; frequency/rate if possible; examples; hypothesized cause; improvement ideas.

Ablation study. At least one ablation is mandatory for Assignments 2 and 3; recommended for Assignment 1 when comparing representations or regularizers.

Limitations and conclusion. Data, method, and experimental limits; generalization; future work; short answer to the research question.

4 Deliverables and reproducibility

4.1 Deliverables per assignment

Unless an assignment chapter adds extras, each assignment requires:

1. Assignment page on GitHub Pages.
2. Source-code repository.
3. Report (following Section 3).
4. Presentation slides.
5. YouTube presentation video.
6. AI Usage Disclosure (landing page + assignment page + report + `AI_USAGE.md`).
7. Checkpoint(s) or reconstruction instructions.
8. LMS submission according to course rules.

Assignments 2 and 3 additionally require the Dataset Proposal record and approval status.

The shared landing page must link to all deliverables for Assignments 1–3 as they become available.

4.2 Reproducibility requirements

The repository must include:

- `README.md` with installation, dataset preparation, train, and evaluate commands.
- Configuration files.
- Seed settings.
- Dependency versions.
- Hardware information.
- Checkpoint access or reconstruction instructions.
- Links to the report and assignment page.
- `AI_USAGE.md`.

Main reported results must be traceable to model configuration, dataset split, checkpoint, log/experiment ID, and the corresponding commit/tag. Submitting only a pre-run notebook without reproduction instructions is not acceptable.

5 Academic integrity and AI-use policy

AI tool use is allowed only when disclosed and verifiable.

5.1 Required disclosure locations

1. AI Usage Disclosure on the shared landing page.
2. AI Usage Disclosure on each assignment page and in each report.
3. A repository file such as `AI_USAGE.md` with detailed logs.

5.2 Mandatory fields per AI tool

For each tool used, record:

- Tool name and version/model if known.
- Member who used it.
- Time or development stage.
- Purpose.

- Affected assignment section(s).
- Representative prompt example or prompt-log link.
- How AI output was edited and verified.
- Member responsible for final verification.
- Sources used for verification (docs/papers/tests).

Suggested categories include literature search, concept explanation, architecture or experiment design, coding assistance, debugging, test generation, report writing, grammar checks, figure or slide generation, and result analysis.

5.3 Disclosure example

```
Tool: [Tool/model name]
Used by: [Member name]
Task: Debugging the validation loop in train.py
Prompt summary: Asked why validation loss was accumulated incorrectly
AI contribution: Suggested sample-weighted loss accumulation
Student verification: Compared with PyTorch docs;
  tested on a small controlled set
Affected files/sections: src/train.py; report Sec. 3.2
Responsible member: [Member name]
```

5.4 If no AI was used

Declare explicitly:

> *The group declares that no generative AI tool was used in this assignment.*

5.5 Student responsibilities

- Students are fully responsible for submitted code, numbers, figures, claims, and narrative.
- Students must understand and be able to explain every part of the submission.
- Do not submit unchecked AI-generated code or text.
- Do not use AI to fabricate data, experimental results, citations, or references.
- Do not claim experiments that were not actually run.
- Do not paste private, restricted, or credential-bearing data into AI tools.

- AI-assisted content must be verified via source code, real runs, official documentation, or scholarly sources.
- Missing or false disclosure may be treated as an academic integrity violation.

The instructor may interview any member and may request explanation, live edits, metric interpretation, partial reproduction, or clarification of student vs. AI-assisted content. Individual contribution may affect the final grade.

6 Grading principles

6.1 Three levels of weighting

1. **Across assignments (course project):** Assignment 1 = 40%, Assignment 2 = 30%, Assignment 3 = 30% (Section 1.2).
2. **Within each assignment (milestones):** official weights in Section 7
   - A1: M1 Draft 25%, M2 Final 75%.
   - A2: M1 Proposal 15%, M2 Draft 25%, M3 Final 60%.
   - A3: M1 Proposal 15%, M2 Draft 25%, M3 Final 60%.
3. **Within each milestone / final package:** content rubrics in Parts II–IV (correctness, understanding, experiments, analysis, reproducibility, presentation, integrity).

6.2 What is assessed

Grading emphasizes correctness, understanding, experimental design, reproducibility, analysis, code quality, report quality, presentation, and academic integrity.

- Core mandatory work dominates the grade of each assignment.
- Extensions cannot replace missing core requirements.
- Do not grade primarily by accuracy.
- Strong numbers without explanation or reproducibility do not receive full credit.
- Late milestones are penalized as in Section 7.2 (20% of the milestone score per week late, up to 7 days).

7 Submission and deadlines

7.1 General rules

- Every milestone below is due at **23:59 Asia/Ho Chi Minh (GMT+7)** on the calendar date shown, unless the LMS states otherwise for that milestone.
- Publish materials on the GitHub Pages site and repository.
- Register the group and submit required files via the course LMS page: `https://lms.hcmut.edu.vn/course/view.php?id=142848`.
- File naming, format, and packaging rules are published on the LMS course page.
- Milestones listed below are **mandatory**. Missing a Dataset Proposal (A2-M1 / A3-M1) blocks main implementation until the proposal is **Approved** or **Approved with conditions**.

7.2 Late-submission policy

- Late submissions are accepted for up to **one week** after the milestone deadline.
- Penalty: **20% of that milestone's score per calendar week late** (or any fraction of a week), applied to the milestone component only.
- Example: an A1-M1 Draft scored 80/100 and submitted 2 days late contributes $0.80 \times 80 = 64$ points toward the M1 component.
- Submissions more than 7 days late for a milestone receive **0** for that milestone unless the instructor grants a written exception in advance.
- Dataset Proposals submitted late still require approval before main work; late penalties still apply to the M1 component.

7.3 Course-wide milestones

Week	Milestone	Due (23:59)	Deliverables (minimum)
1	Handbook released	—	Assignments 1–3 specifications published
3	Group + Pages skeleton	09 Sep 2026	Group registration on LMS; repo; landing page with links to A1/A2/A3

The Group + Pages skeleton is a **mandatory gate** (participation / course-project eligibility). Incomplete registration may delay grading of later milestones.

7.4 Assignment 1 milestones (40% of course project)

Official weights within Assignment 1 (sum to 100% of A1):

Week	Milestone	Due (23:59)	Weight	Of A1
5	M1 Draft	23 Sep 2026	25%	of Assignment 1
9	M2 Final	21 Oct 2026	75%	of Assignment 1

- **M1 Draft (minimum):** EDA; Dataset/DataLoader; train/val loop; **Linear + MLP** runnable (CNN optional).
- **M2 Final (minimum):** all five models; full comparison; report; slides; YouTube video; assignment page; checkpoints.
- LSTM/GRU and Transformer are required in **M2 Final**, not in M1 Draft.

7.5 Assignment 2 milestones (30% of course project)

Official weights within Assignment 2 (sum to 100% of A2):

Week	Milestone	Due (23:59)	Weight	Of A2
7	M1 Proposal	07 Oct 2026	15%	of Assignment 2
10	M2 Draft	28 Oct 2026	25%	of Assignment 2
12	M3 Final	11 Nov 2026	60%	of Assignment 2

- **M1 Proposal:** Dataset Proposal (task, data, split, metrics, baseline, compute estimate). Main experiments may begin only after **Approved** or **Approved with conditions**.
- **M2 Draft:** simple baseline + pretrained model in progress; EDA; preliminary metrics.
- **M3 Final:** controlled experiment; error analysis; report; slides; video; assignment page; checkpoints.

7.6 Assignment 3 milestones (30% of course project)

Official weights within Assignment 3 (sum to 100% of A3):

Week	Milestone	Due (23:59)	Weight	Of A3
13	M1 Proposal	18 Nov 2026	15%	of Assignment 3
14	M2 Draft	25 Nov 2026	25%	of Assignment 3
15	M3 Final	02 Dec 2026	60%	of Assignment 3

- **M1 Proposal:** Dataset Proposal (modalities, pair validity, split unit, metrics, baselines). Main experiments may begin only after **Approved** or **Approved with conditions**.
- **M2 Draft:** unimodal A, unimodal B, simple fusion with quantitative results.
- **M3 Final:** improved fusion; ablation; report; slides; video; assignment page; checkpoints.

7.7 Milestone notation

- **M1, M2, M3** = Milestone 1, 2, 3 *within* the named assignment (not midterm exams).
- **Draft / Proposal** milestones are graded progress checkpoints; **Final** milestones require all deliverables in Section 4 for that assignment.
- Draft milestones may accept partial reports and work-in-progress repositories; Final milestones must satisfy the full rubric in Parts II–IV.
- Content rubrics in Parts II–IV describe *what* is assessed; milestone weights above describe *when* that score is collected.

7.8 Calendar overview

Date (23:59 GMT+7)	**Assignment**	**Milestone**
09 Sep 2026	Course	Group + Pages skeleton
23 Sep 2026	A1	M1 Draft (25% of A1)
07 Oct 2026	A2	M1 Proposal (15% of A2)
21 Oct 2026	A1	M2 Final (75% of A1)
28 Oct 2026	A2	M2 Draft (25% of A2)
11 Nov 2026	A2	M3 Final (60% of A2)
18 Nov 2026	A3	M1 Proposal (15% of A3)
25 Nov 2026	A3	M2 Draft (25% of A3)
02 Dec 2026	A3	M3 Final (60% of A3)

This handbook and the LMS course page are the student-facing sources of truth. If an LMS announcement for a specific milestone differs from this handbook, follow the LMS announcement for that milestone.

Part II

Assignment 1

8 Assignment 1 overview

Title: Foundations of Deep Learning Pipelines and Architectures.
Topic: From Linear Models to Modern Sequence Models: A Comparative Study for Image Classification.
Course-project weight: 40% (see Section 1.2).

Students implement an end-to-end deep learning pipeline for a controlled-scale image classification problem, then build and compare multiple architecture families under a shared data split and evaluation protocol.

8.1 Scope

- Exploratory data analysis (EDA).
- Train/validation/test splitting.
- Dataset and DataLoader design.
- Preprocessing and augmentation.
- Training/validation/test pipelines in PyTorch.
- Implementation and comparison of architecture families.
- Evaluation, error analysis, and a reproducible report.

Required core work must be completed fully. Extensions are optional and *cannot* replace missing core requirements. General website, report, deliverable, AI, and submission rules are in Part I. Milestone deadlines for Assignment 1: Section 7.4 (M1 Draft 25%: 23 Sep 2026; M2 Final 75%: 21 Oct 2026; due 23:59 GMT+7).

9 Assignment 1 learning objectives

Upon completing this assignment, students will be able to:

1. Construct a complete PyTorch training pipeline for image classification, including Dataset, DataLoader, preprocessing, augmentation, optimization, and evaluation.
2. Implement and explain linear/softmax, MLP, CNN, LSTM/GRU, and Transformer models for the same classification task.

3. Compare models using multiple quantitative metrics and qualitative evidence, not accuracy alone.
4. Analyze inductive bias and data representation choices (flattened pixels, spatial features, row/column/patch sequences).
5. Produce reproducible code, reports, GitHub Pages materials, and a presentation video that satisfy the course deliverable standard.

10 Assignment 1 dataset requirements

- **MNIST** may be used for development and debugging only.
- **Primary results** must be reported on **Fashion-MNIST**.
- **CIFAR-10** may be used as an *optional extension*.
- All models in the main comparison must use the **same data split** (same train/validation/test partitions and the same random seed).
- Report the official or custom split, the seed, stratification policy (if any), and the exact number of samples per split.

Groups must state clearly which dataset is used for debugging, which dataset is used for the main comparison, and which (if any) extension datasets were evaluated.

11 Assignment 1 technical requirements

11.1 Mandatory models

Students must implement, train, and compare **all** of the following:

1. **Linear / softmax classifier**
    - Flatten each image into a vector.
    - Use a linear layer to produce logits.
    - Use cross-entropy loss.
    - Do *not* apply softmax before `CrossEntropyLoss`.
2. **Multilayer Perceptron (MLP)**
    - At least one hidden layer.
    - Explain activation functions and any regularization used.

3. **Convolutional Neural Network (CNN)**
   - Design the architecture yourselves.
   - Do *not* only call a pretrained model as the sole CNN submission.
   - Explain convolution, pooling, and feature maps.
4. **LSTM or GRU**
   - Represent each image as a sequence of rows, columns, or patches.
   - Explain timestep definition, input size, and hidden representation.
5. **Transformer**
   - Represent each image as rows/columns or patches.
   - Include token embedding/projection and positional encoding.
   - Explain attention inputs and outputs.

11.2 Optional extensions

Groups may choose one or more of the following (extensions do not replace core models):

- Compare LSTM with GRU.
- Mamba or another state-space model (*not* mandatory).
- CIFAR-10 experiments.
- Robustness evaluation.
- Interpretability analysis.
- Compare a self-implemented PyTorch pipeline with Keras or another high-level framework.
- Compare models under a matched parameter or compute budget.

11.3 Framework notes

- Primary framework: **PyTorch**.
- Code must be a reproducible project (not only a one-off notebook without instructions).
- Pretrained backbones are allowed only as extensions or ablation references; the five mandatory models must be implemented and trained for the main comparison.

12 Assignment 1 experimental requirements

12.1 Mandatory comparison outputs

- Accuracy.
- Macro-F1.
- Number of parameters.
- Training time.
- Inference time.
- Training and validation curves.
- Confusion matrix.
- Correct and incorrect prediction examples.
- Analysis of how data representation and inductive bias affect performance.

Do **not** conclude that one model is better based on accuracy alone.

12.2 Fairness constraints

- Same dataset split for all main models.
- Same evaluation protocol and metrics.
- Report hardware, software versions, seed(s), and checkpoint selection rule.

13 Assignment 1 report focus

Follow the three-part structure in Section 3. For this assignment, EDA must include at least class distribution, input size, imbalance analysis, and representative samples.

14 Assignment 1 within-assignment rubric

This rubric sums to 100% *within Assignment 1.* The assignment then contributes 40% of the course-project grade.

Component	What is assessed	Weight
Problem & data understanding	Formulation, EDA, split, leakage awareness	15%
Data & training pipeline	Dataset/DataLoader, preprocessing, train/-val/test loop	15%
Methods & implementation	Five mandatory models, correct losses, explanations	30%
Experimental design & results	Fair comparison, metrics beyond accuracy, curves/tables	20%
Analysis & error analysis	Inductive bias discussion, failure cases, limitations	10%
Reproducibility, report & presentation	Repo, pages, video, disclosure, writing quality	10%

Part III

Assignment 2

15 Assignment 2 overview

Title: Deep Learning on Large-Scale Data and Specialized Tasks.
Topic: Deep Learning on Large-Scale Data for Specialized Computer Vision or Language Tasks.
Course-project weight: 30% (see Section 1.2).

Students reuse and extend the pipeline skills from Assignment 1 on a larger and more specialized problem. Each group works deeply on **one** approved track; groups are not required to cover all tracks.

15.1 Scope

- Select or be assigned one specialized task track.
- Submit a Dataset Proposal and wait for approval before full implementation.
- Build a simple baseline and at least one pretrained/modern model with a clear fine-tuning protocol.
- Run at least one controlled experiment, quantitative and qualitative evaluation, error analysis, and compute-cost analysis.

Required core work must be completed fully. Extensions cannot replace missing core requirements. General website, report, deliverable, AI, and submission rules are in Part I. Milestone deadlines for Assignment 2: Section 7.5 (M1 Proposal 15%: 07 Oct; M2 Draft 25%: 28 Oct; M3 Final 60%: 11 Nov 2026; due 23:59 GMT+7).

16 Assignment 2 learning objectives

Upon completing this assignment, students will be able to:

1. Propose, justify, and obtain approval for a dataset and task protocol suitable for specialized deep learning work.
2. Adapt an end-to-end pipeline to larger-scale data and task-specific annotations/metrics.
3. Compare a simple baseline with at least one pretrained or modern model under a documented fine-tuning strategy.
4. Design and interpret at least one controlled ablation or factor experiment.
5. Analyze errors, compute cost, and reproducibility for a specialized computer vision or language task.

17 Assignment 2 task tracks

Each group chooses or is assigned one of the following:

- Image classification.
- Text classification or sequence labeling.
- Object detection.
- Semantic or instance segmentation.
- Person/vehicle re-identification.
- Monocular depth estimation.
- Another specialized task approved by the instructor.

Depth of work on one track is required; breadth across all tracks is not.

18 Assignment 2 dataset proposal and approval

Before implementation, each group must submit a **Dataset Proposal** containing:

- Dataset name, source, version, and license.
- Task, input, and output definitions.
- Number of samples and annotation types.
- Preliminary distribution analysis.
- Data-splitting plan.
- Split unit used to prevent leakage.
- Evaluation metric(s).
- Baseline plan.
- Planned pretrained model(s).
- Compute estimate.
- Subset selection rules, if any.

A dataset may be used for the main assignment **only after approval**. Approval status values: Approved; Approved with conditions; Rejected.

19 Assignment 2 dataset requirements

Default thresholds below apply unless an exception is explicitly approved. Exceptions require evidence of task difficulty, academic value, or special annotation/compute constraints.

19.1 Image classification

- At least 5 classes.
- At least 5,000 training samples.
- Enough per-class samples for meaningful evaluation.
- Do **not** use MNIST or Fashion-MNIST.

19.2 Text classification

- At least 5 classes.
- At least 5,000 training samples.
- Check duplicates, text length distribution, and class imbalance.
- Describe tokenization and normalization.

19.3 Object detection

- Default: at least 5 classes.
- About 3,000 or more training images.
- About 5,000 or more object instances.
- Analyze bounding-box size distribution and objects-per-image statistics.

19.4 Segmentation

- Default: at least 3 foreground classes (excluding background).
- About 1,000 or more images with masks.
- Binary datasets only if they have real practical difficulty and are approved.
- Analyze per-class pixel distribution.

19.5 Re-identification

- About 300 or more training identities.
- Multiple images per identity.
- Prefer multiple cameras/contexts.
- Test identities must not overlap training identities.
- Use a correct query/gallery protocol.

19.6 Depth estimation

- About 5,000 or more RGB–depth pairs, unless separately approved.
- Describe depth scale, invalid pixels, and scene split.
- Do not place the same scene in both train and test when that would cause leakage.

20 Assignment 2 technical requirements

Each group must provide:

1. A simple baseline.
2. At least one pretrained or modern model.
3. A clear fine-tuning procedure.
4. At least one controlled experiment, such as:
   - Freeze vs. full fine-tune.
   - With vs. without augmentation.
   - Backbone A vs. B.
   - Loss A vs. B.
   - Resolution A vs. B.
   - Another task-appropriate controlled factor.
5. Quantitative evaluation.
6. Qualitative evaluation.
7. Error analysis.
8. Compute-cost analysis.

20.1 Task-specific metrics

- Classification: accuracy and macro-F1.
- Detection: mAP and AP by class and/or size.
- Segmentation: mIoU and Dice; instance segmentation also reports mask AP.
- Re-ID: Rank-1, Rank-5, and mAP.
- Depth: Abs Rel, RMSE, and δ accuracy.
- Text tasks: task-appropriate metrics with explicit justification.

21 Assignment 2 experimental requirements

For every reported experiment, state hypothesis, changed factor, fixed factors, evaluation metric(s), and decision criterion. At least one ablation (or equivalent controlled factor study) is mandatory. Report hardware, software versions, seeds, checkpoint selection rules, training time, and inference cost for main configurations.

22 Assignment 2 report focus

Follow Section 3. Task-specific EDA expectations:

- Classification: class distribution, input size/length, imbalance.
- Detection: object counts, class distribution, bounding-box sizes.
- Segmentation: pixel distribution, region size, mask quality.
- Re-ID: identity distribution, images per identity, camera distribution.
- Depth: depth distribution, invalid pixels, scene distribution.

23 Assignment 2 within-assignment rubric

This rubric sums to 100% *within Assignment 2*. The assignment then contributes 30% of the course-project grade.

Component	What is assessed	Weight
Problem & data understanding	Proposal quality, EDA, split/leakage control	15%
Data & training pipeline	Large-scale loading, preprocessing, training loop	15%
Methods & implementation	Baseline, pretrained/modern model, fine-tuning	25%
Experimental design & results	Controlled experiment, metrics, compute cost	20%
Analysis & error analysis	Qualitative evidence, error taxonomy, ablation	15%
Reproducibility, report & presentation	Repo, pages, video, disclosure, writing	10%

Part IV

Assignment 3

24 Assignment 3 overview

Title: Multimodal Deep Learning.
Topic: Multimodal Deep Learning: Representation, Fusion, and Evaluation.
Course-project weight: 30% (see Section 1.2).

Students design, train, and evaluate a system that uses **at least two modalities** that are genuinely related (same entity, event, scene, or timestamp).

24.1 Scope

- Choose an approved multimodal task.
- Submit a Dataset Proposal and obtain approval before full implementation (same process as Assignment 2).
- Build unimodal baselines, a simple fusion baseline, and an improved multimodal/fusion model.
- Analyze when modalities help each other, when they conflict, and what ablations reveal.

Required core work must be completed fully. Multimodal results are *not* required to beat unimodal baselines; if they do not, groups must explain why with evidence. General website, report, deliverable, AI, and submission rules are in Part I. Milestone deadlines for Assignment 3: Section 7.6 (M1 Proposal 15%: 18 Nov; M2 Draft 25%: 25 Nov; M3 Final 60%: 02 Dec 2026; due 23:59 GMT+7).

25 Assignment 3 learning objectives

Upon completing this assignment, students will be able to:

1. Formulate a multimodal learning problem with genuine cross-modal alignment.
2. Implement modality-specific encoders and at least one fusion strategy (early, intermediate, or late).
3. Compare unimodal baselines against simple and improved multimodal models using task-appropriate metrics.
4. Conduct ablations that isolate modality contribution and fusion design choices.
5. Analyze complementary and conflicting modality evidence, errors, limitations, and compute cost.

26 Assignment 3 task options

Groups may choose one of the following (or another instructor-approved multimodal task):

- Image–text classification.

- Image–text retrieval.
- Visual question answering (VQA).
- Image captioning.
- Open-vocabulary detection.
- Text-guided segmentation.
- RGB–depth prediction.
- Video–text retrieval.
- Re-identification with image and metadata/text.

27 Assignment 3 dataset proposal and approval

Use the same Dataset Proposal process as Assignment 2. Include:

- Dataset name, source, version, and license.
- Task, modalities, input, and output.
- Number of paired samples and annotation types.
- Preliminary pair-quality and distribution analysis.
- Split plan and leakage-prevention unit (entity/event/scene when needed).
- Metrics, baselines, planned models, compute estimate, and subset rules.

Approval statuses: Approved; Approved with conditions; Rejected. Implementation of the main experiments may begin only after approval.

28 Assignment 3 dataset requirements

28.1 Cross-modal validity constraints

- Every sample must contain at least two modalities linked by the same entity, event, scene, or timestamp.
- Do **not** randomly pair data from two independent datasets.
- Check missing, noisy, and mismatched pairs.
- Split by entity/event/scene when needed to prevent leakage.
- If one entity has multiple records, keep all of that entity's records in the same split.

28.2 Default size thresholds

- Multimodal classification: at least 5 classes and about 5,000 paired samples.
- Image–text retrieval: about 5,000 or more image–text pairs.
- Captioning: about 5,000 or more images; prefer multiple captions per image.
- VQA: about 10,000 or more question–image pairs.
- RGB–depth: about 5,000 or more aligned pairs.
- Video–text: a lower threshold may be approved when processing cost is high.

Exceptions require instructor approval with justification.

29 Assignment 3 technical requirements

29.1 Mandatory baselines and models

Each group must build and compare:

1. Baseline using modality A only.
2. Baseline using modality B only.
3. A simple fusion baseline.
4. An improved multimodal model or improved fusion design.

Multimodal performance is not required to exceed unimodal performance. If it does not, analyze causes using data properties and experimental evidence.

29.2 Mandatory method content

- Encoder for each modality.
- Representation of each modality.
- Data alignment procedure.
- Fusion stage (early, intermediate, or late).
- Fusion operation.
- Per-modality losses and the overall loss.
- Training procedure.

- Missing-modality handling, if applicable.
- Pretraining and fine-tuning strategy.
- At least one ablation that removes or changes a modality or fusion component.

29.3 Reference metrics by task

- Classification: accuracy and macro-F1.
- Retrieval: Recall@1, Recall@5, Recall@10, and median rank.
- Captioning: CIDEr/BLEU plus qualitative evaluation.
- VQA: accuracy under the stated protocol.
- Detection/segmentation: mAP or mIoU as appropriate.
- RGB–depth: corresponding depth metrics.

30 Assignment 3 experimental requirements

Mandatory result content:

- Unimodal vs. multimodal comparison.
- Quantitative evaluation with task metrics.
- Qualitative examples.
- Cases where modalities support each other.
- Cases where modalities conflict.
- Error analysis.
- Ablation study.
- Compute-cost analysis.

For each experiment, state hypothesis, changed factor, fixed factors, metrics, and conclusion criteria. Report hardware, software versions, seeds, checkpoint rules, and training/inference cost for main configurations.

31 Assignment 3 report focus

Follow Section 3. Multimodal EDA must cover pair counts, missing modality rates, text length if applicable, and pair quality. Pipeline diagrams may use:

```
Raw paired data → preprocessing → data loader
→ encoders → fusion → loss
→ optimization → prediction → evaluation
```

32 Assignment 3 within-assignment rubric

This rubric sums to 100% *within Assignment 3*. The assignment then contributes 30% of the course-project grade.

Component	**What is assessed**	**Weight**
Problem & data understanding	Pair validity, EDA, leakage-aware splits	15%
Data & training pipeline	Multimodal loading, alignment, training loop	15%
Methods & implementation	Unimodal baselines, fusion, improved multimodal model	25%
Experimental design & results	Comparisons, metrics, compute cost	20%
Analysis & error analysis	Support/conflict cases, ablation, limitations	15%
Reproducibility, report & presentation	Repo, pages, video, disclosure, writing	10%